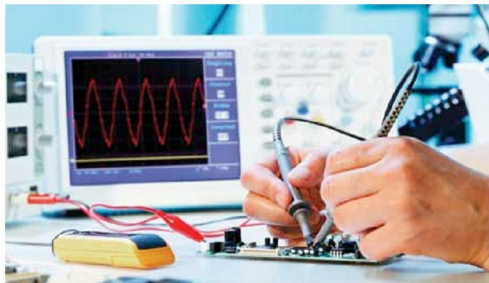


Analogue Design: A Challenging But Satisfying Option

Designers who envision, plan, execute and inspire are the lifeline of an organisation. EFY caught up with experts from various companies, who shared their views on career in analogue design in India. This article focuses on the current scenario, scope, entry-level roles, skills required and industry experts' suggestions



ABHISHEK MUTHA

With good understanding and skills in areas such as custom circuit layout, electrical design and simulation, analogue designers plan, organise, execute and document as they steer their organisation towards its goals. The Indian electronics industry looks for designers who are flexible across domains and technologies.

"Especially, an expert in analogue design is highly valued. Lots of analogue design experience and strong basics in the field can fetch high-profile jobs in India," says P. Chow Reddy, manager-R&D (power division), ICOMM Tele Ltd.

Analogue design scenario in India

Basavaraj Garadi, chief expert, Robert Bosch Engineering and Business Solutions, feels that the analogue design

scenario is not that great. He says, "Sadly, India is not where analogue designs thrive. Analogue circuit design is not what excites the hardware designers in India, perhaps because it is not well understood. Designers would rather work on high-speed digital boards, clocking at mind-boggling multi-gigahertz speeds and churning out thousands of MIPS of processing."

Garadi adds, "Besides, over the years, engineers have resorted to processing analogue signals in digital domain, by converting analogue signals to digital format and converting back to analogue after processing. Forgotten are the days when the only computers available were analogue. Even amplification of signals, which was typically perceived as pure analogue processing, is now being realised in binary format, with Class D amplifiers using power devices (MOSFETs) driven as switches rather than in linear mode."

Engineers have realised the potential of digital processing of analogue signals. "Also working in favour of digital processing is the rapid fall in prices of digital signal processors. Therefore pure analogue board design will not gain any further foothold in India," says Garadi. However, he points out, "If a digital electronic device is to interact with the real world, it will always need an analogue interface. Therefore in their minimal form, analogue circuit designs would perhaps be restricted to front-end interfaces. This is as far as board designs are concerned. However, IC or ASIC designs offer a different picture, providing a great deal of promise for analogue designers, both in terms of excitement as well as salaries."

Comparing the Indian scenario with the Japanese electronics industry, Reddy says, "The Japanese electronics industry looks at specialisation and expertise in a single field. For example, if a designer starts working on small-signal amplifiers and earns experience in it, he retires from this field only and never concentrates on any other field. They never come across words like 'job security.' This is not the scenario in India."

Sharing his thoughts on requirements in analogue sector, Binu Raj S., AVP-engineering, Dexcel Electronics Designs, says, "We as a design service company get less than 15 per cent of the requirement in analogue domain. It is a scarce but specialised domain."

On a positive note, Rajani Rao, manager (analogue and mixed-signal IP development), LSI India Research Development, says, "Indeed, these are exciting times for analogue design sector in India."